

Compulsory Task 1

1. A model that allocates which mail folder an email should be sent to (work, friends, promotions, important), like Gmail's inbox tabs.

Category: Text classification

Reasoning: This model needs to analyse the content of incoming/received emails and assign them to predefined categories, such as work, important, promotion, and spam. Thus, this is a text classification-based mode that makes use of machine learning to label text based on its contents.

Example: An email/gmail could be classified as a promotion if it contains common words or phrases associated with marketing or sales, such as 'Offer', 'Buy one, get one free', 'Subscribe', 'Savings', etc. (MailGenius, n.d.).

2. A model that helps decide what grade to award to an essay question. This can be used by a university professor who grades a lot of classes or essay competitions

Category: Text classification and sentiment analysis

Reasoning: This model needs to evaluate the quality of a written essay and assign a grade. This could involve both text classification and sentiment analysis to assess the tone and coherence of the essay.

Example: Let's consider an automated essay grading system that needs to grade essays on scientific concepts. Research done by Dennis *et al.*, 2025, found that this type of grading system could give a better score to essays that contain relevant content, organisation and coherence, language use (in this case appropriate scientific terms relevant to the chosen topic), and sentiment analysis (in this case, to evaluate the formal and objective tone throughout the essay). Since it is a scientific paper, it requires neutral tones and could penalise personal opinions or informal languages or tones.

Thus, text classification and sentiment analysis fit this model well.

3. A model that provides assistive technology for doctors to provide their diagnosis. Remember, doctors ask questions, so the model will use the patients' answers to provide probable diagnoses for the doctor to weigh and make decisions on

Category: Question answering

Reasoning: This model could help doctors diagnose patients by analysing their responses to relevant medical questions and then suggesting possible diagnoses. This fits a question-answering model well since it interprets user inputs, such as patient answers, and then provides an appropriate response.

Example: IBM Watson Health and Babylon Health utilise AI-powered question-answering models that assist doctors in diagnosing patients (IBM Watson Health, 2021; Babylon Health, 2020) A patient could input their symptoms, such as dry cough, fever, shortness of breath, and heart palpitations. The AI would then process these inputs using NLP and identify the key medical terms and check known diseases/ailments with these symptoms. If unsure, it could ask follow-up questions, like 'Have you been in an area with high malaria/scarlet fever/yellow fever?'. It could then generate a list of possible conditions and rank them based on responses. This, however, won't replace a doctor's own diagnosis but could suggest going to see a doctor or specialist based on urgency.

Thus, this model fits the question-answer category.

References

N.d. refers to no date provided, found, or supplied on the webpage.

- MailGenius (n.d.) *What is the Promotions tab in Gmail?* Available at: <https://www.mailgenius.com/what-is-the-promotions-tab-in-gmail> (accessed: 3 March 2025).
- Dennis C. Gabon, Albert A. Vinluan, Jennifer T. Carpio. (Year). 'Automated Grading of Essay Using Natural Language Processing: A Comparative Analysis with Human Raters Across Multiple Essay Types', *Jisem Journal*, 10(6), 2468-4376. Available at: <https://www.jisem-journal.com/index.php/journal/article/view/700/244> (Accessed: 3 March 2025).
- IBM Watson Health. (2021). *Watson for Oncology: Improving treatment decisions with AI*. Journal of Medical Internet Research. Available at: <https://cancer.jmir.org/2021/4/e27850/> [Accessed 4 Mar. 2025].
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